

PiEEG

OctoBio Kit

User Manual

Brain rhythms in. Real-world actions out.

- Brain-to-hardware learning kit
- 16-channel EEG + Arduino starter kit
- Programmed with blocks in the browser

26	16	15
Lessons	EEG channels	Built-in modules

octobridge.pieeg.com

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Introduction

Welcome to the **OctoBio Kit**, a brain-to-hardware learning kit from **PiEEG**. You will read real biosignals from your own head, build programs with blocks, and make lights, motors, sounds and a screen respond.

What is the OctoBio Kit?

Octopus 16
16-channel EEG board
by PiEEG

+

All-in-One Starter Kit for Arduino
sensor & actuator board
by Elecrow (SEA0004T)

=

OctoBio Kit
both together, with the octoBridge web lab

The starter board is designed and made by **Elecrow**. PiEEG supplies the Octopus 16 EEG board, the OctoBio firmware for both boards, and the octoBridge web lab that links them. Together we call them the **OctoBio Kit**.

Everything runs in one browser tab at **octobridge.pieeg.com**. There is no Arduino IDE to install and no code to compile: you stack blocks and press Run.

This manual has two parts. **Part A** (Lessons 1–20) uses only the starter board, so you can start without a headset. **Part B** (Lessons 21–26) adds the Octopus 16 and your brain signals.

Safety first

Read this before anyone puts the Octopus 16 on.

> **Not a medical device.** The OctoBio Kit hasn't been certified by any regulator for use with the human body. It is a learning and experimentation tool.

1. **Battery only while worn.** Wear the Octopus 16 on a 5 V power bank only, never while it's plugged into a computer or a wall charger. The board has no isolation from its power supply.
2. **Never connect the relay to mains voltage.**
3. **No diagnosis.** Focus and relax are uncalibrated learning indicators. Never use them for diagnosis, or as a measure of mood or ability.
4. **Implants.** Don't use it with a pacemaker or other implanted device without asking a physician.
5. **Skin.** Stop if the skin becomes irritated.
6. **Pins.** Press the pogo pins straight down and keep them clean with isopropyl alcohol. No oil or grease.

Power source	While worn
5 V power bank	✔ Safe: no path to the mains
Laptop USB	✘ Only for flashing, board off the head
Wall charger	✘ Never

What's in the box

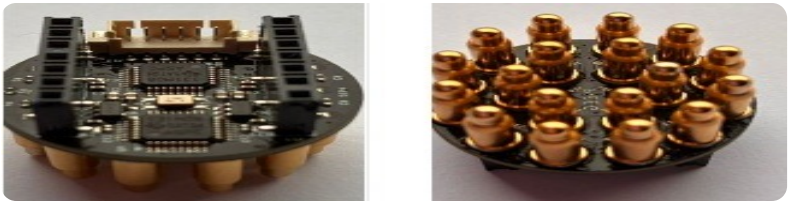
Included

1 × Octopus 16 EEG board	16 dry pogo-pin electrodes, 2 × ADS131M08, Seeed Studio XIAO ESP32-S3 with BLE antenna. *By PiEEG.*
1 × All-in-One Starter Kit for Arduino	SEA0004T, Arduino Uno compatible, 15 modules in one case. *By Elecrow.*
Software	OctoBio firmware and the octoBridge web lab

SKU PIA24028P · 570 g · Sold by PiEEG on Elecrow

You also need

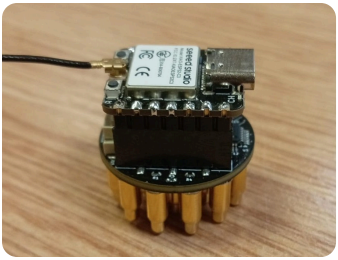
5 V power bank	Small, for the Octopus 16 while it is worn
Computer	With Bluetooth, running **desktop Chrome or Edge**
USB data cables	USB-C for the XIAO, one for the starter board. Charge-only cables can't flash or stream.



Octopus 16: top (controller socket) and underside (16 gold pogo-pin electrodes)

Meet the Octopus 16

The brain side · by PiEEG



A round 16-channel EEG board. Its spring-loaded **pogo pins are the electrodes**, so there's no gel, no cap and no cables.

A Seeed Studio XIAO ESP32-S3 on top streams the signal over **Bluetooth LE**. The browser turns it into band power.

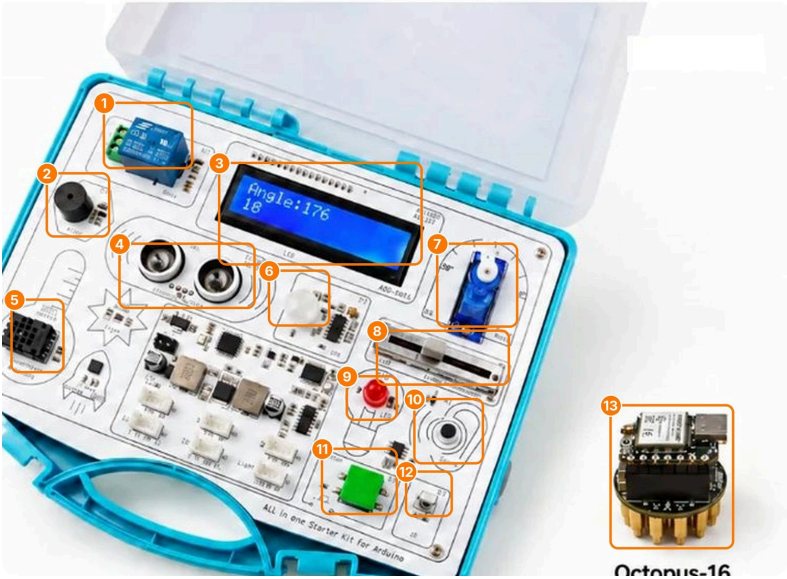
What it reads

Band power	δ delta 0.5–4 Hz · θ theta 4–8 Hz · α alpha 8–13 Hz · β beta 13–30 Hz · γ gamma 30+ Hz (relative)
Focus	beta ÷ (beta + theta), uncalibrated
Relax	alpha ÷ (alpha + beta), uncalibrated
RMS	Signal size in µV
EEG ok	Green when RMS is between 2 and 200 µV

Electrodes	16 dry pogo pins	ADC	2 × ADS131M08, 24-bit
Controller	XIAO ESP32-S3	Rate	~250 SPS
Name	`Octopus-16-XXXX`		

Meet the starter board

The room side · All-in-One Starter Kit for Arduino by Elecrow



1	Relay	8	Slide potentiometer ("knob")
2	Buzzer	9	Red LED (D10)
3	16×2 LCD screen	10	Sound sensor
4	Ultrasonic distance sensor	11	Button
5	Temperature & humidity (DHT20)	12	IR receiver
6	PIR motion sensor	13	Octopus 16 (PiEEG)
7	Servo motor		

Also on the board: light sensor (BH1750), 6-axis motion sensor, second light D8, and a connector for the soil moisture sensor. The board is Arduino Uno compatible (ATmega328P) and talks to octoBridge over USB at 115200 baud.

Modules, pins and values

You never need pin numbers in octoBridge: every module has its own block. They are listed here for reference and for the Arduino lessons in Elecrow's manual.

Module	Connection	Value in octoBridge
Red LED	D10	light D10: on / off / flash
Second light	D8	light D8: on / off
Button	D7	pressed / released
Relay	D4	on / off (no mains!)
Servo	D9	0–180°
IR receiver	D2	remote key code
Ultrasonic	D6 trig, D5 echo	distance, cm
Slide potentiometer	A0	knob, raw 0–1023
Sound sensor	A1	raw 0–1023
PIR motion	A2	motion / no motion
Soil moisture	A3	raw 0–1023
Buzzer	digital pin	on / off
Light sensor BH1750	I²C	lux
Temp & humidity DHT20	I²C	°C, %
6-axis motion sensor	I²C	tilt (no heading)
16×2 LCD	I²C	2 lines × 16 characters

Raw counts (0–1023) are not calibrated units: they tell you more or less, not how much.

Getting started

Step 1 · Open the lab

Go to **octobridge.pieeg.com** in **desktop Chrome or Edge**. The lab needs Web Bluetooth and Web Serial, which only those browsers have. Safari, Firefox and phones can't open USB serial.

No account, no install. The Bluetooth and USB connections stay in your browser tab; nothing is uploaded to a PiEEG server.

Page	What it does
Home	What you can read, what you can move, ready mixes
Flash	Put firmware on the boards. First time only.
1 · Connect	Pair the Octopus 16 and/or the starter board
2 · Board	Live numbers, lights, motor and screen
3 · Blocks	Stack a program and run it
4 · Learn	What a reading is allowed to mean
5 · Write	The same program, written like a page
6 · Repeat	Six ready machines to run and remix

Getting started

Step 2 · Flash the firmware once

Open **Flash**, plug each board in with a USB **data** cable and install its firmware from the browser. No Arduino IDE. You only do this the first time.

Octopus 16

1. Plug the XIAO ESP32-S3 into the computer (board **off the head**).
2. Click **Install Octopus-16** and choose its port.
3. No port? Hold **BOOT**, tap **RESET**, release BOOT and install again.
4. Unplug and replug when it finishes.

> It now shows up over Bluetooth as `Octopus-16-XXXX`.

Starter board

1. Close any other serial monitor (Arduino IDE, terminal).
2. Plug in the starter board and click **Flash octoBio Kit**.
3. Choose its port. Flashing takes about 20 seconds.

> The screen says **octoBio Kit Ready**.

Works with Arduino Uno / ATmega328P boards (official 16U2 or CH340).

Getting started

Step 3 · Connect both devices

Open **Connect**. Both devices stay live at once. Skip a device you don't have: its numbers stay still until it's connected.



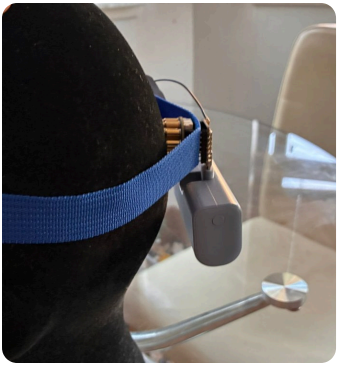
1. Power the Octopus 16 from the **power bank**, then choose `Octopus-16-XXXX` in the Bluetooth picker.
2. Plug the starter board into the computer and choose its port in the serial picker.
3. When the port opens, the board restarts. Wait until its status says **live**.

The board answers at 115200 baud: `INF protocol=1 ready`, and the **Board** page starts moving.

Tip: for Part A you only need the starter board. Leave the Octopus 16 for Part B.

Getting started

Step 4 · Put on the Octopus 16



Strap the Octopus 16 over the **back of the head** (the visual cortex) so alpha shows up clearly.

Press it **straight down** so every pin touches the scalp. Part the hair under the pins if needed.

> **Battery only.** While it's on a head, the Octopus 16 runs from the power bank only, never plugged into a computer or a wall charger.

Getting started

Step 5 · Check the signal

On **Board**, look for the **EEG ok** check: RMS between 2 and 200 µV. Then run the quick tests. If they behave as described, you're reading real biosignals.

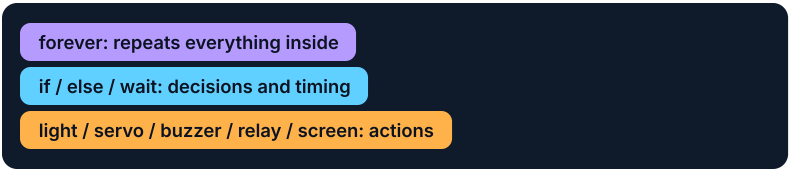
Test	What you should see
EEG ok is green	RMS between 2 and 200 µV
Close your eyes for 10 s	Alpha rises, then drops when the eyes open
Blink and clench the jaw	RMS and gamma jump, then settle
Sit still for the first run	Movement shifts the pins

> A blink or a jaw clench raises RMS and gamma. That's muscle, not a thought, and a good first lesson.

Programming with blocks

Open **Blocks**, stack blocks and press **Run**. The program runs live and sends commands to the board, so there's no compiling and no second sketch to flash.

Block colours in this manual



One condition. One action.

Every lesson is a single rule, so you see cause and effect straight away. The numbers in the lessons (50 lux, 300, 35 %…) are starting values: check your own readings on the **Board** page and adjust.

Five block levels

Level	Name	
1	Story	Primary school
2	Words	
3	Bands · δ θ α β γ	
4	Measure · numbers and units	
5	Raw · raw values	University

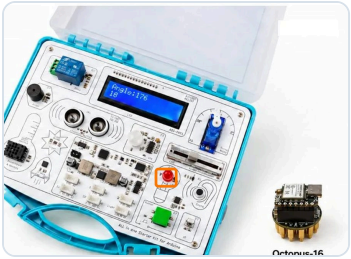
Same kit, same lessons. Move up a level as the class does. The blocks in this manual are written in plain words; the exact look depends on the level you pick.

PART A · STARTER BOARD ONLY

Lesson 1 · LED Blink

Introduction

Your first program. Turn the red LED on and off once per second, forever.



HARDWARE USED

Red LED (D10)

YOU'LL LEARN

Output · Loop · Timing

How it works

An LED (light-emitting diode) glows when current flows through it in the right direction. The board switches pin D10 on and off, and the **wait** block keeps each state for a set time.

Block program



What you'll see

The red LED blinks: one second on, one second off.

Try this

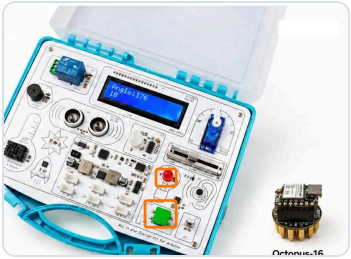
- Change both waits to 0.2 s. What happens?
- Try the **flash** action instead of on/off.

PART A · STARTER BOARD ONLY

Lesson 2 · Button Controls the LED

Introduction

Press the green button and the LED turns on. Let go and it turns off.



HARDWARE USED

Button (D7) · Red LED (D10)

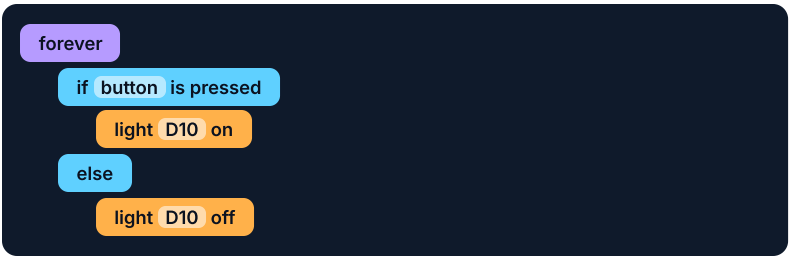
YOU'LL LEARN

Digital input · If / else

How it works

A button is a digital input: it is either pressed or not pressed. The **if / else** block checks it many times a second and picks one of two actions.

Block program



What you'll see

The LED follows the button.

Try this

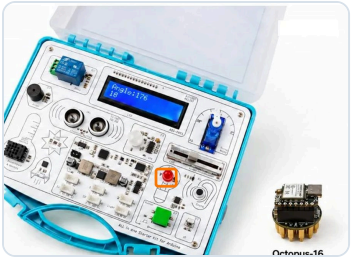
- Swap the actions so the LED is on until you press.
- Add the buzzer to the pressed branch.

PART A · STARTER BOARD ONLY

Lesson 3 · Two Lights

Introduction

Make the two lights take turns, like a railway crossing.



HARDWARE USED

Light D10 · Light D8

YOU'LL LEARN

Sequence · Two outputs

How it works

Two outputs controlled from one loop. The order of the blocks decides what happens first. (The breathing-light lesson in the Elecrow manual needs Arduino code; in octoBridge the lights are on, off or flash.)

Block program



What you'll see

The two lights blink one after the other.

Try this

- Make the pattern faster, then slower.
- Add a beep from the buzzer on every change.

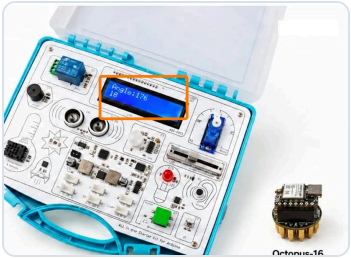
New lesson (replaces Elecrow Lesson 3 · Breathing Light)

PART A · STARTER BOARD ONLY

Lesson 4 · Hello, Screen

Introduction

Write your own message on the 16×2 screen.



HARDWARE USED

16×2 LCD screen (I²C)

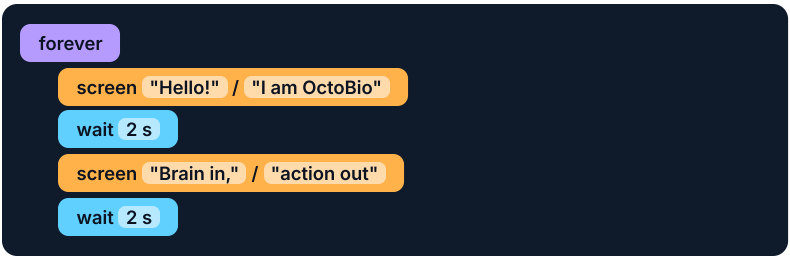
YOU'LL LEARN

Text output · 16×2 display

How it works

The LCD1602 screen shows 2 lines of 16 characters. Each **screen** block sets both lines. Longer text is cut at 16 characters.

Block program



What you'll see

The screen swaps between two messages every 2 seconds.

Try this

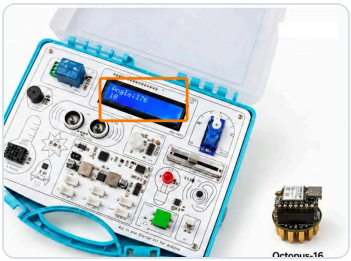
- Write your name on line 1.
- Count how many characters fit on one line.

PART A · STARTER BOARD ONLY

Lesson 5 · Moisture Monitor

Introduction

Show the soil moisture reading on the screen.



HARDWARE USED

Soil moisture sensor (A3) · 16×2 LCD

YOU'LL LEARN

Analog input · Raw counts

How it works

The soil sensor gives an analog signal. The board turns it into a raw count from 0 to 1023. A raw count is not a percentage: it only tells you *more* or *less* wet.

Block program



What you'll see

The number changes when you put the probe in dry and in wet soil.

Try this

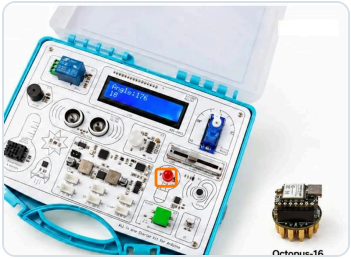
- Write down the number in dry soil and in a glass of water.
- Use **map** to turn 0–1023 into 0–100.

PART A · STARTER BOARD ONLY

Lesson 6 · Smart Street Light

Introduction

When it gets dark, the light turns on by itself.



HARDWARE USED

Light sensor (BH1750, I²C) · Red LED (D10)

YOU'LL LEARN

Threshold · Units: lux

How it works

The BH1750 measures light in lux. Indoor light is usually a few hundred lux; covering the sensor with a hand drops it close to zero. A **threshold** splits the reading into dark and bright.

Block program



What you'll see

Cover the light sensor with your hand and the LED turns on.

Try this

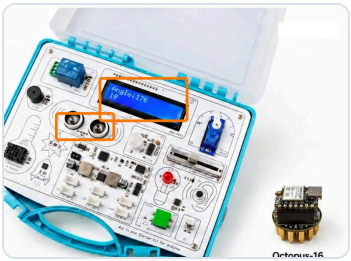
- Check your room's lux on the **Board** page and pick a better threshold.
- Use **flash** when it is very dark.

PART A · STARTER BOARD ONLY

Lesson 7 · Distance Display

Introduction

Measure the distance to your hand and show it on the screen.



HARDWARE USED

- Ultrasonic sensor (trig D6, echo D5)
- 16×2 LCD

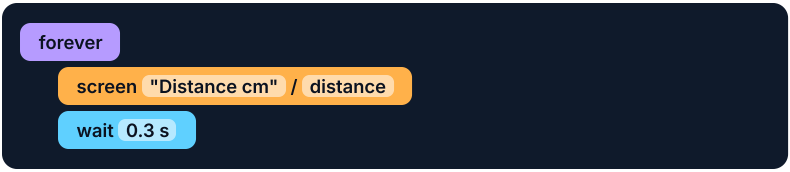
YOU'LL LEARN

- Sensors and units
- Echo timing

How it works

The sensor sends a sound pulse above 20 kHz and times the echo. Distance = speed of sound × time ÷ 2. It needs a surface facing it: a missed echo gives a **negative** number, not "very far".

Block program



What you'll see

Move your hand closer and further: the number follows.

Try this

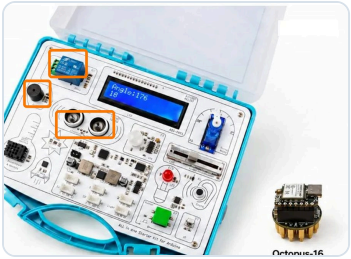
- What is the shortest and longest distance it can read?
- Try a soft surface like a pillow.

PART A · STARTER BOARD ONLY

Lesson 8 · Obstacle Alarm

Introduction

If something comes too close, sound the alarm and click the relay.



HARDWARE USED

Ultrasonic sensor · Buzzer · Relay (D4)

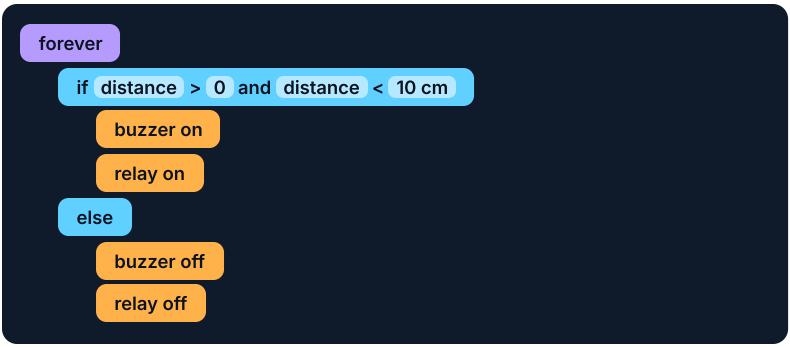
YOU'LL LEARN

AND logic · Relay

How it works

The relay is a switch moved by an electromagnet; you can hear it click. In this kit it only switches low-voltage things. **Never connect the relay to mains voltage.**

Block program



What you'll see

Bring your hand closer than 10 cm: beep and click.

Try this

- Why do we check **distance > 0** too?
- Change the alarm distance to 20 cm.

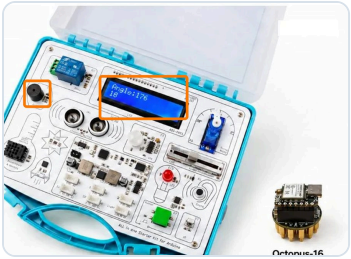
Elecrow Lesson 8 · Obstacle Close Range Alarm

PART A · STARTER BOARD ONLY

Lesson 9 · Plant Watering Reminder

Introduction

The kit reminds you when the soil is dry.



HARDWARE USED

Soil moisture sensor · Buzzer · 16×2 LCD

YOU'LL LEARN

Threshold · Two actions

How it works

One reading, two actions. Below the line, the soil is dry: show a message and beep. Find your own line by testing dry and wet soil in Lesson 5.

Block program

```
forever
    if soil < 300
        screen "Water me!"
        buzzer on
    else
        screen "Soil is fine"
        buzzer off
```

What you'll see

Pull the probe out of the soil: the screen asks for water.

Try this

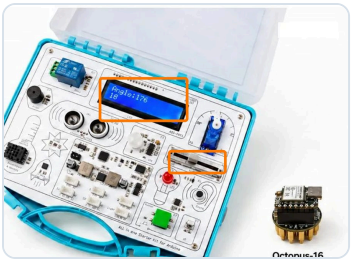
- Make the buzzer beep (on, wait, off) instead of a long tone.
- Add light D10 as a silent warning.

PART A · STARTER BOARD ONLY

Lesson 10 · Slider Display

Introduction

Read the slide potentiometer and show its value.



HARDWARE USED

- Slide potentiometer (A0), the "knob" in octoBridge
- 16×2 LCD

YOU'LL LEARN

- Analog input
- Mapping

How it works

A potentiometer is a variable resistor. Sliding it changes the voltage on A0, which the board reads as 0 to 1023.

Block program



What you'll see

Slide it from end to end: the number goes from about 0 to about 1023.

Try this

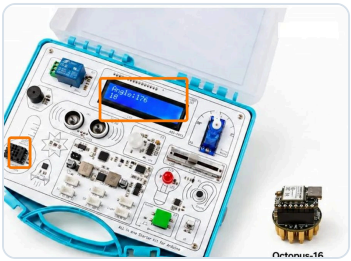
- Show the value as 0–100 % with **map**.
- Turn light D10 on when the knob is past halfway.

PART A · STARTER BOARD ONLY

Lesson 11 · Temperature & Humidity

Introduction

Build a small weather station on the screen.



HARDWARE USED

- Temperature & humidity sensor (DHT20, I²C)
- 16×2 LCD

YOU'LL LEARN

- I²C sensors
- Units: °C, %

How it works

The DHT20 measures temperature (°C) and relative humidity (%). It talks to the board over I²C, a two-wire bus that the light sensor, motion sensor and screen share.

Block program



What you'll see

The screen shows the temperature, then the humidity.

Try this

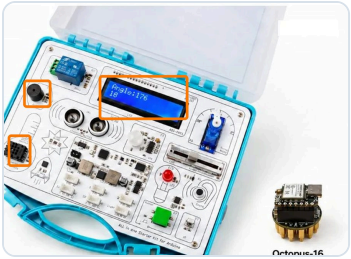
- Breathe gently on the sensor. Which value changes most?
- Put a cold drink next to it.

PART A · STARTER BOARD ONLY

Lesson 12 · Weather Reminder

Introduction

Warn when it is too hot.



HARDWARE USED

DHT20 sensor · Buzzer · 16×2 LCD

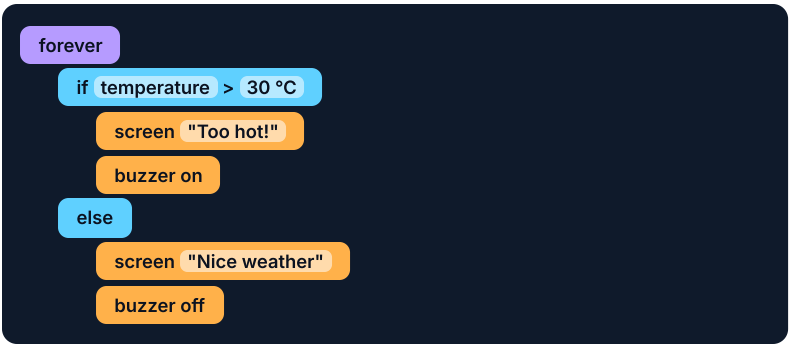
YOU'LL LEARN

Comparing values · Units

How it works

A rule can check a measured value against a limit. Here: above 30 °C, it is too hot.

Block program



What you'll see

Warm the sensor with your fingers to trigger the warning.

Try this

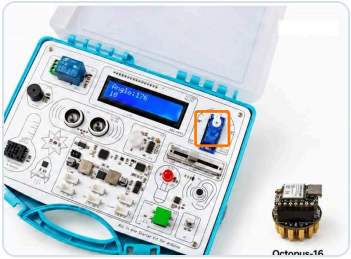
- Add a humidity rule: above 70 % show "Take umbrella".
- Lower the limit to 2 °C above room temperature.

PART A · STARTER BOARD ONLY

Lesson 13 · Servo Sweep

Introduction

Move the servo arm to set angles.



HARDWARE USED

Servo motor (D9)

YOU'LL LEARN

Actuator · Angles

How it works

A servo is a motor with a built-in position sensor. You tell it an angle from 0 to 180° and it goes there and holds.

Block program



What you'll see

The arm steps from 0° to 90° to 180°, then jumps back.

Try this

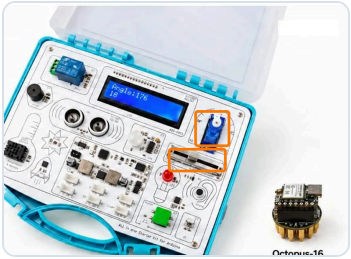
- Add more steps (45°, 135°) for a smoother sweep.
- Make it a windscreen wiper.

PART A · STARTER BOARD ONLY

Lesson 14 · Follow the Knob

Introduction

A warm-up with no headset. Slide the knob and the servo follows.



HARDWARE USED

Slide potentiometer (A0) | Servo motor (D9)

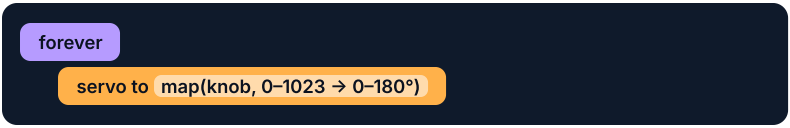
YOU'LL LEARN

Mapping a range | Analog input

How it works

The knob is a raw count from 0 to 1023. The **map** block turns that range into 0 to 180 degrees. The same map block works for any reading, including brain signals later.

Block program



What you'll see

Analog in, angle out: the arm follows your hand.

Try this

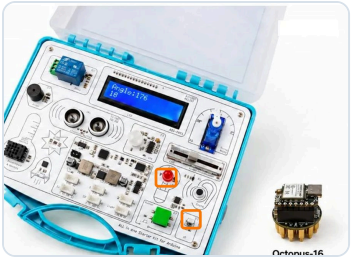
- Reverse it: map 0–1023 to 180–0°.
- Use only half the range: 0–1023 → 45–135°.

PART A · STARTER BOARD ONLY

Lesson 15 · IR Remote Light

Introduction

Use the infrared remote to switch the light.



HARDWARE USED

IR receiver (D2) · IR remote · Red LED (D10)

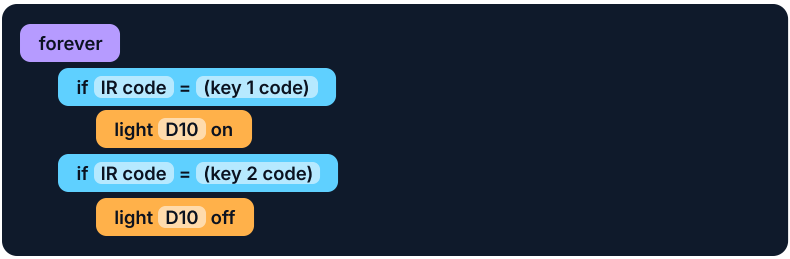
YOU'LL LEARN

Codes · Wireless input

How it works

An IR remote flashes invisible infrared light in a pattern. Each key sends its own code. Open the **Board** page, press a key and note the code it shows.

Block program



What you'll see

Key 1 turns the light on, key 2 turns it off.

Try this

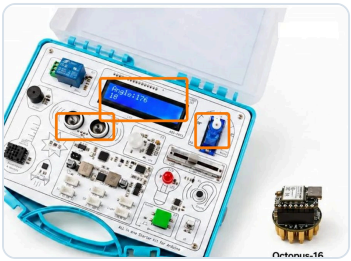
- Use a third key to make it flash.
- Point the remote away. How far does it work?

PART A · STARTER BOARD ONLY

Lesson 16 · Polite Automatic Door

Introduction

When someone comes close, the door opens and says hello.



HARDWARE USED

Ultrasonic sensor · Servo motor (the door) · 16×2 LCD

YOU'LL LEARN

One condition, two actions

How it works

The servo arm is the door. The distance sensor sees a visitor. One condition drives two actions at once.

Block program

```
forever
    if distance > 0 and distance < 15 cm
        servo to 90°
        screen "Welcome!"
    else
        servo to 0°
        screen "Door closed"
```

What you'll see

Wave your hand in front of the sensor: the door opens.

Try this

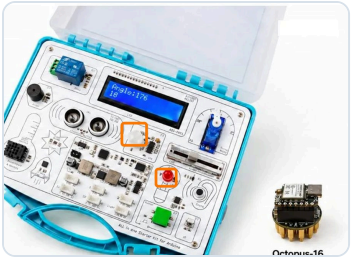
- Keep the door open for 3 s after the visitor leaves.
- Use the button as a doorbell.

PART A · STARTER BOARD ONLY

Lesson 17 · Motion Night Light

Introduction

The light turns on when someone moves nearby.



HARDWARE USED

PIR motion sensor (A2) · Red LED (D10)

YOU'LL LEARN

Motion sensing · Digital input

How it works

A PIR (passive infrared) sensor sees changes in the heat that people give off. It reports *motion* or *no motion*. It needs a few seconds to settle after power-up.

Block program



What you'll see

Walk past the kit: the light turns on.

Try this

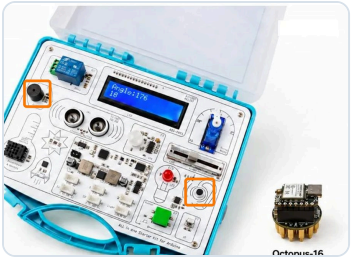
- How long does the sensor stay "on" after you stop?
- Add a screen message: "Someone is here".

PART A · STARTER BOARD ONLY

Lesson 18 · Sound Reminder

Introduction

Clap your hands and the buzzer answers.



HARDWARE USED

Sound sensor (A1) · Buzzer

YOU'LL LEARN

Raw counts · Threshold

How it works

The microphone turns sound into a voltage. The board reads it as a raw count from 0 to 1023. Loud sounds make bigger numbers.

Block program



What you'll see

Clap near the sound sensor: a short beep.

Try this

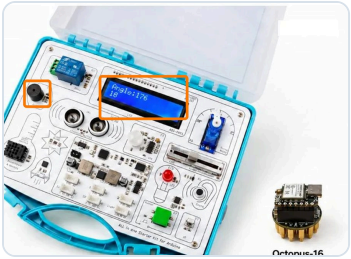
- Find the quiet-room number on the Board page and set a better threshold.
- Use a clap to switch the light instead.

PART A · STARTER BOARD ONLY

Lesson 19 · Tilt Alarm

Introduction

Tilt the case and the kit complains.



HARDWARE USED

6-axis motion sensor (I²C) · Buzzer · 16×2 LCD

YOU'LL LEARN

Axes · Motion

How it works

The motion sensor measures acceleration and rotation on three axes. octoBridge turns it into tilt. It has no compass, so it cannot tell which way is north.

Block program



What you'll see

Lift one side of the case: the alarm sounds.

Try this

- Watch the tilt values on the Board page while you tilt each way.
- Make a spirit level with the servo as the needle.

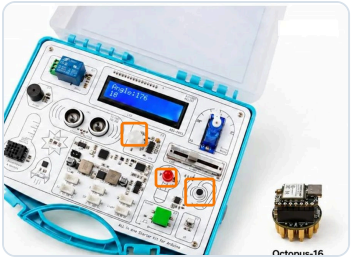
Elecrow Lesson 19 · Calculation of Acceleration

PART A · STARTER BOARD ONLY

Lesson 20 · Smart Corridor Light

Introduction

The corridor light turns on only when it is dark **and** someone is there.



HARDWARE USED

PIR motion sensor · Light sensor · Red LED (D10)

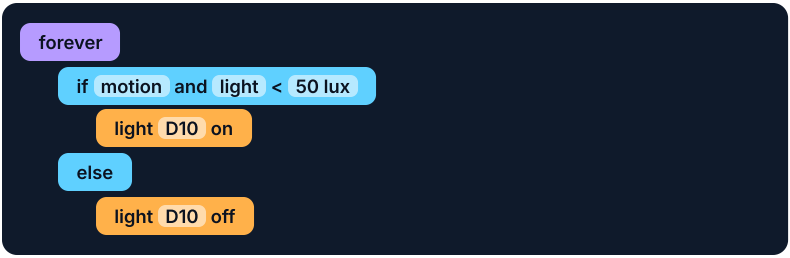
YOU'LL LEARN

AND / OR logic · Two sensors

How it works

Two conditions joined with **and**: both must be true. This saves energy, because the light never turns on in daylight.

Block program



What you'll see

Cover the light sensor and wave: the light turns on. Uncover it: no light.

Try this

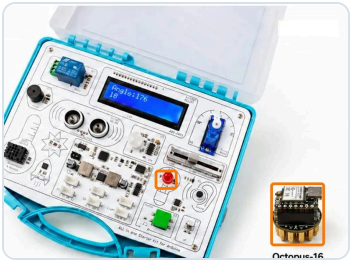
- Add **or** *sound* > 600 so a clap also works.
- Show "Night mode" on the screen when it is dark.

PART B · BRAIN + BOARD

Lesson 21 · Eyes Closed, Light On

Introduction

Close your eyes and the alpha rhythm (8 to 13 Hz) over the back of the head grows. One rule turns that into a light.



HARDWARE USED

- Octopus 16 (worn, on battery)
- Red LED (D10)

YOU'LL LEARN

- Biology: alpha rhythm
- If / else
- Threshold

How it works

When you close your eyes, the visual cortex at the back of the head "idles" and produces a strong alpha rhythm. octoBridge shows alpha as a share of the total EEG power.

Block program



What you'll see

Alpha is high with eyes closed, so the light is on. Open your eyes and it drops.

Try this

- Find your own alpha number with eyes open and closed.
- Count to 10 with eyes closed: how fast does it react?

OctoBio brain lesson

PART B · BRAIN + BOARD

Lesson 22 · Focus Meter

Introduction

The focus index is mapped onto the servo angle, so the arm becomes a needle.



HARDWARE USED

Octopus 16 · Servo motor

YOU'LL LEARN

Mapping a range · Beta band · Servo

How it works

Focus = beta ÷ (beta + theta). It is an **uncalibrated** indicator: it means "higher than a moment ago", not a score. Never use it to grade a person.

Block program



What you'll see

Do mental arithmetic and watch the needle; then relax.

Try this

- Compare reading a book with looking out of the window.
- Smooth the needle: add a short wait.

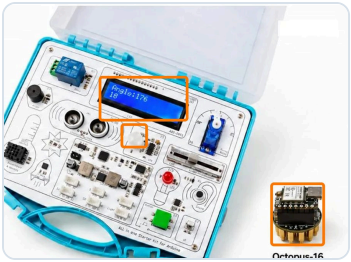
OctoBio brain lesson

PART B · BRAIN + BOARD

Lesson 23 · Quiet Room

Introduction

Two readings, one action. If alpha is high and the motion sensor sees nobody, the screen asks for quiet.



HARDWARE USED

Octopus 16 · PIR motion sensor · 16×2 LCD

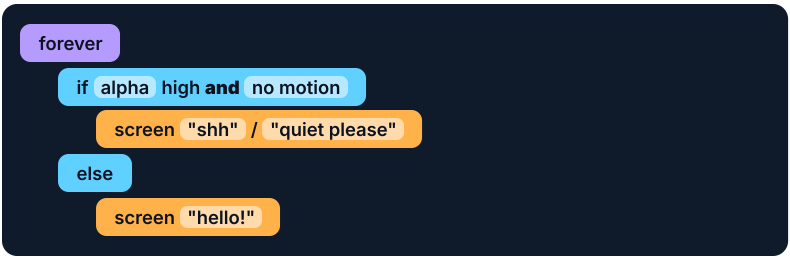
YOU'LL LEARN

Combining two signals · PIR · LCD

How it works

This mixes a brain signal with a room signal. Both conditions must be true.

Block program



What you'll see

Close your eyes while nobody walks past: "shh" appears.

Try this

- Ask a friend to walk past while your eyes are closed.
- Add light D8 as a "do not disturb" sign.

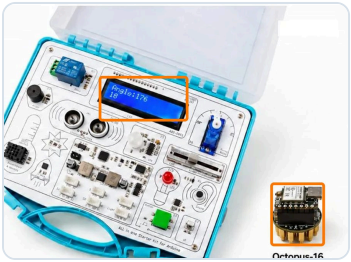
OctoBio brain lesson

PART B · BRAIN + BOARD

Lesson 24 · Blink Is Not a Thought

Introduction

Blinks and jaw clenches are muscle, and they show up in the EEG as big swings in RMS and gamma. This rule flags them.



HARDWARE USED

Octopus 16 · 16×2 LCD

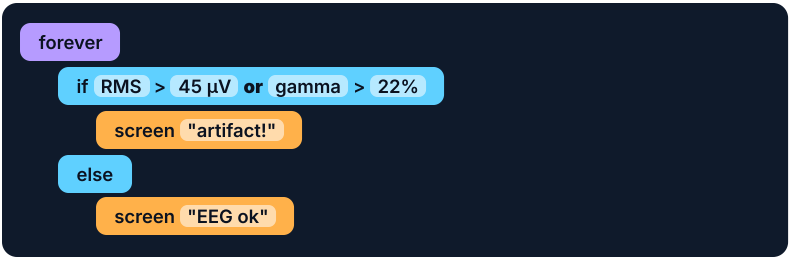
YOU'LL LEARN

Biology: artifacts · Signal quality · OR logic

How it works

Muscles near the electrodes make signals much bigger than brain waves. Good scientists learn to spot these **artifacts** before they trust a reading.

Block program



What you'll see

Blink hard or clench your jaw: "artifact!". Sit still: "EEG ok".

Try this

- Which makes the biggest artifact: blinking, chewing or talking?
- Why could an artifact fool the Focus Meter?

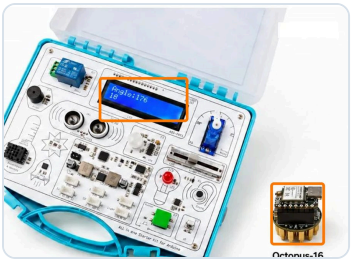
OctoBio brain lesson

PART B · BRAIN + BOARD

Lesson 25 · Brain Dashboard

Introduction

Show your live brain numbers on the kit's screen.



HARDWARE USED

Octopus 16 | 16×2 LCD

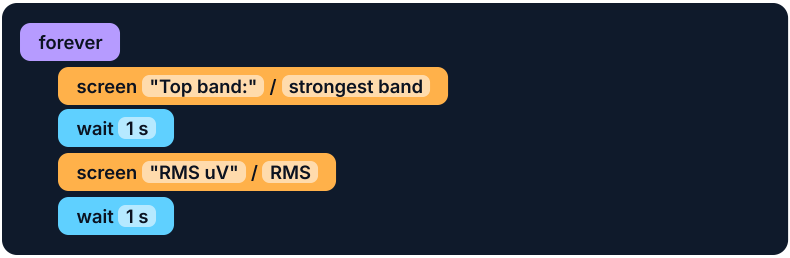
YOU'LL LEARN

Band power | Units: µV

How it works

The strongest band is the band with the most power right now. RMS tells you the overall size of the signal in microvolts (µV).

Block program



What you'll see

The strongest band changes when you close your eyes, talk or move.

Try this

- Which band wins most often with eyes open?
- Show *relax* instead of RMS.

OctoBio brain lesson

PART B · BRAIN + BOARD

Lesson 26 · Mind Door

Introduction

Open the servo "door" by closing your eyes.



HARDWARE USED

Octopus 16 | Servo motor

YOU'LL LEARN

Brain + actuator | Reliable signals

How it works

The same rule as Lesson 21, but with a moving action. A clear, strong signal (alpha with eyes closed) works much better than a weak one (focus).

Block program



What you'll see

Eyes closed: the door opens. Eyes open: it closes.

Try this

- Combine with Lesson 16: open only if someone is close **and** your eyes are closed.
- Add a "door open" message.

OctoBio brain lesson

Write and Repeat

Write

The **Write** page shows the same program written like a page of text. Use it to read a program aloud, copy it into a notebook, or move from blocks towards real code.

Repeat

The **Repeat** page has six ready machines to run and remix: eyes closed → light on, walk-by → buzzer, knob → servo, focus → servo, distance → light, and alpha + no motion → a line on the screen. Running one does not save over your own stack.

Learn

The **Learn** page explains what a reading is allowed to mean, for example why focus is not a score and why a raw count is not a percentage. It does not send commands to the board.

Teaching tips

- Start every class with **Follow the knob** (Lesson 14): no headset needed.
- Teach **Blink Is Not a Thought** (Lesson 24) early.
- Clean the pogo pins between students.

Troubleshooting

Problem	Fix
The lab won't connect	Use desktop Chrome or Edge on the HTTPS site.
No serial port	Use a USB **data** cable. Close the Arduino IDE and other serial monitors.
Board numbers don't move	The board restarts when the port opens. Wait for **live**.
XIAO port not found	Hold BOOT, tap RESET, release BOOT, try again.
Octopus silent after flashing	Unplug and replug the USB once.
Octopus on Bluetooth but no EEG, LED double-blinks	The XIAO is seated the wrong way round. Reseat it so the 5V / GND / 3V3 labels line up.
EEG ok stays red	Press straight down so every pin touches the scalp. Sit still.
Distance is negative	The echo was missed. Point the sensor at a facing surface.
A pogo pin is stuck	A drop of IPA in the barrel, gently work the plunger, let it dry. No oil or grease.

Specifications

Octopus 16 EEG board · PiEEG

Channels	16 EEG, dry spring-loaded pogo pins
ADC	2 × Texas Instruments ADS131M08, 24-bit
Controller	Seeed Studio XIAO ESP32-S3
Wireless	Bluetooth LE, ~250 SPS
Power	5 V power bank, battery only while worn

All-in-One Starter Kit for Arduino · Elecrow SEA0004T

Board	Arduino Uno compatible (ATmega328P)
Connection	USB serial, 115200 baud
Sensors	Ultrasonic distance, light (BH1750), temperature & humidity (DHT20), PIR motion, sound, soil moisture, slide potentiometer, 6-axis motion, button, IR receiver
Outputs	LED, second LED, relay, buzzer, servo, 16×2 LCD
Firmware	OctoBio firmware, flashed once from the browser

octoBridge web lab · PiEEG

Platform	Chrome or Edge (desktop), octobridge.pieeg.com
Connection	Web Bluetooth + Web Serial, both live at once
Programming	Blocks, 5 levels, runs live without compiling
EEG values	Relative band power δ θ α β γ, focus, relax, signal quality

Going further: Arduino code

The starter board is a standard Arduino Uno-compatible board, so you can also program it in C++ with the **Arduino IDE**. Elecrow's own *All-in-one Starter Kit for Arduino User Manual* has 21 Arduino code lessons for the same modules, and its example code is on Elecrow's GitHub.

> **Uploading an Arduino sketch replaces the OctoBio firmware.** octoBridge will not see the board until you flash it again: open **Flash** in octoBridge and click **Flash octoBio Kit**. It takes about 20 seconds.

A good path for older students

1. Build the rule with blocks in octoBridge.
2. Read it on the **Write** page.
3. Write the same rule as an Arduino sketch, using the pin table on page 8.
4. Flash OctoBio back when you're done.

The Octopus 16 can also be used in your own web apps with the PiEEG JavaScript SDK (`octopus-16`), and on cloud.pieeg.com.

OctoBio Kit

Brain rhythms in. Real-world actions out.

octobridge.pieeg.com

Support

Support for the OctoBio software, firmware and the Octopus 16 comes from the PiEEG team:

Discord · discord.gg/5Ps24p47
GitHub · github.com/pieeg-club
Email · pieeg@pieeg.com

The All-in-One Starter Kit for Arduino (SEA0004T) is designed and manufactured by Elecrow. Octopus 16, the OctoBio firmware and octoBridge are by PiEEG. The OctoBio Kit is not a medical device.

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